Supplementary Information: How temperature, photoperiod, and evolutionary history shape forest leafout

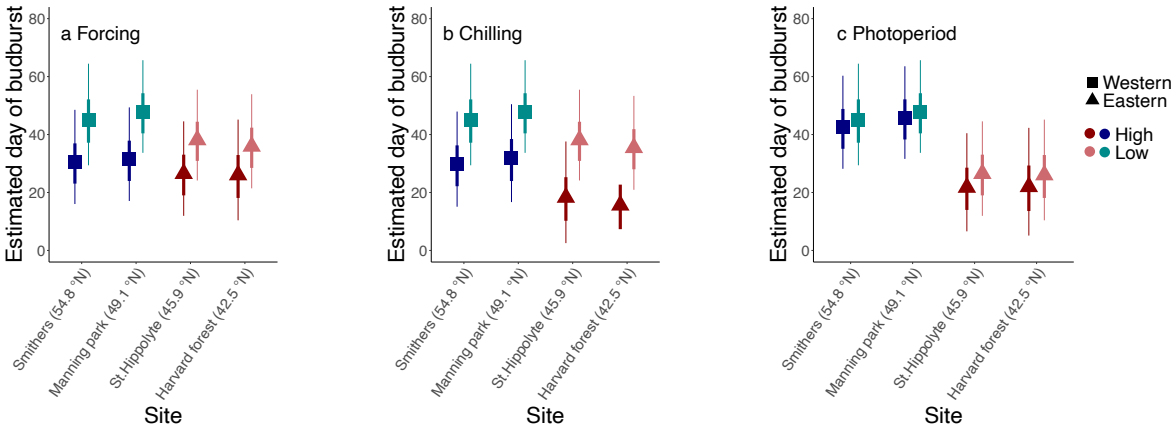


Figure S1: Estimated day of budburst in response to forcing (**a**) across sites under low chilling and short photoperiods, chilling across sites under low forcing and short photoperiods (**b**), and across photoperiods (**c**) under low forcing and chilling for species sampled from our four sites. The thin bars represent the 90% uncertainty interval, while the thicker bars represent the 50% uncertainty interval, and symbols the mean.

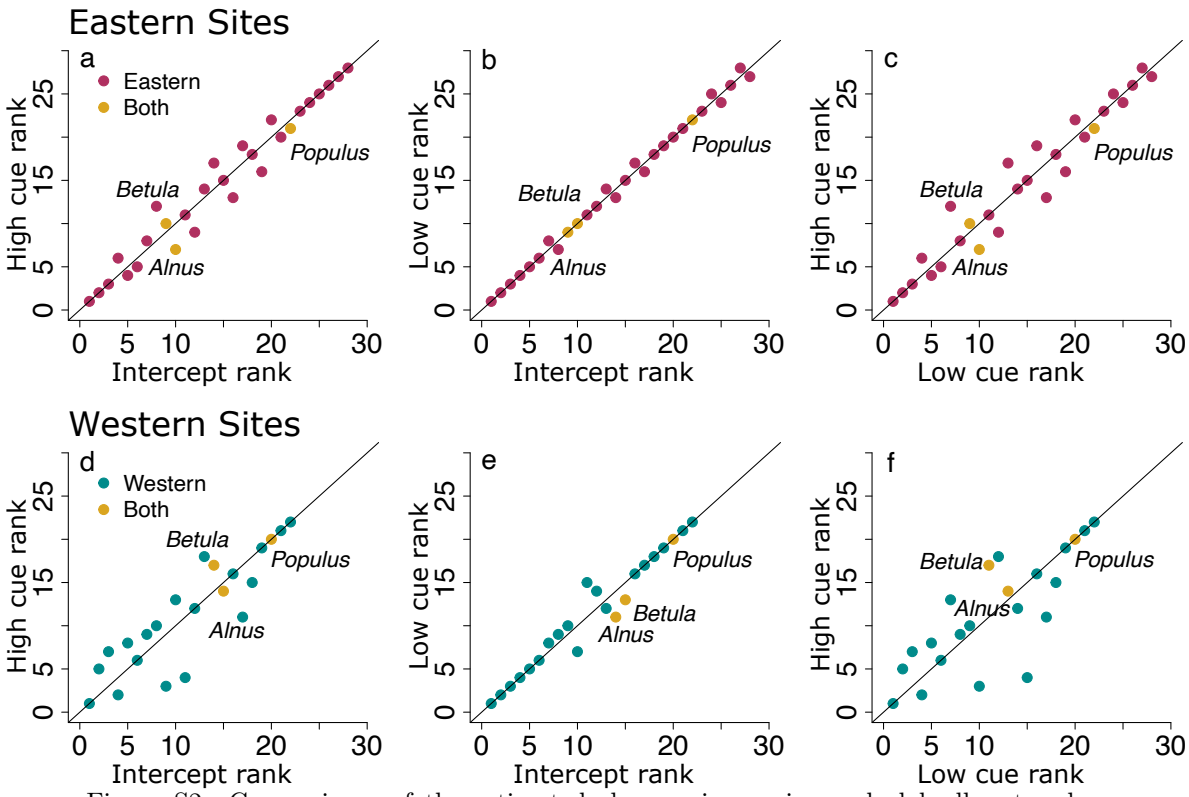


Figure S2: Comparisons of the estimated changes in species ranked budburst order, between species-level effects (species intercept) and under high cues (**a, d**), species-level effects and under low cues (**b, e**), and under low versus high cues (**c, f**), for our eastern species (in red, **a-c**) and western species (in blue, **d-f**). For the three species that occur in both transect (shown in yellow), *Alnus incana* exhibited the greatest rank change with a difference of three, while *Betula papyrifera* had a rank difference of two and *Populus tremuloides* a rank difference of one.

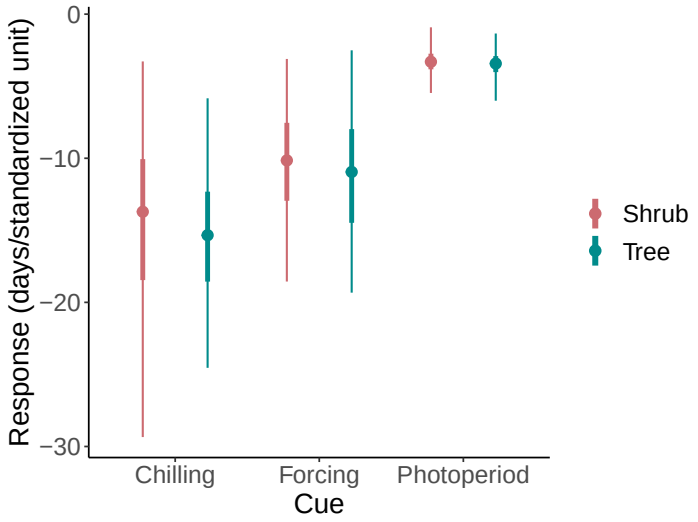


Figure S3: Comparisons of posterior distributions for cue estimates between shrub and tree species. Circles represent the mean, while the thicker error bars represent the 50% uncertainty interval, and thinner line the 90% quantile interval.

Table S1: Mean day of budburst following the start of forcing treatments, estimated from raw observational data of 47 species at our two western sites, E.C. Manning park and Smithers B.C., Canada, and our two eastern sites, Harvard Forest, MA, USA and St. Hippolyte, QC, Canada.

Species	Spp. abbreviation	Harvard forest	St. Hippoltye	Manning park	Smithers
Acer glabrum	acegla			36	39
Acer pensylvanicum	acepen	16	18		
Acer rubrum	acerub	22	25		
Acer saccharum	acesac	45	36		
Alnus incana	alninc			28	30
Alnus incana	alninc	33	25		
Alnus viridis	alnvir			44	43
Amelanchier alnifolia	amealn			19	18
Aronia melanocarpa	aromel	14			
Betula alleghaniensis	betall	20	21		
Betula lenta	betlen	30			
Betula papyrifera	betpap				31
Betula papyrifera	betpap	17	18		
Corylus cornuta	corcor	25	19		
Cornus stolonifera	corsto			15	17
Fagus grandifolia	faggra	42	43		
Fraxinus nigra	franig	38	38		
Hamamelis virginiana	hamvir	44			
Ilex mucronata	ilemuc	16	15		
Kalmia angustifolia	kalang	30	32		
Lonicera canadensis	loncan	17	16		
Lonicera involucrata	loninv			22	20
Lyonia ligustrina	lyolig	31			
Menziesia ferruginea	menfer			43	46
Nyssa sylvatica	nyssyl	32			
Populus balsamifera	popbal			30	31
Populus grandidentata	popgra	33	31		
Populus tremuloides	poptre			46	35
Prunus pensylvanica	prupen	18	16		
Quercus alba	quealb	45			
Quercus rubra	querub	36	34		
Quercus velutina	quevel	52			
Rhamnus frangula	rhafran	32			
Rhododendron albiflorum	rhoalb			19	
Rhododendron prinophyllum	rhopri	29			
Ribes lacustre	riblac			29	23
Rubus parviflorus	rubpar			28	30
Sambucus racemosa	samrac			33	
Shepherdia canadensis	shecan			25	24
Sorbus scopulina	sorsco			21	19
Spiraea alba	spialb	18	20		
Spiraea betulifolia	spibet			24	18
Spiraea pyramidata	spipyr			26	22
Symphoricarpos albus	symalb			27	32
Vaccinium membranaceum	vacmem			22	23
Vaccinium myrtilloides	vacmyr	13	17		
Viburnum cassinoides	vibcas	15	18		
Viburnum edule	vibedu			19	8
Viburnum lantanoides	viblan	31	28		

Table S2: Approximate chill units from our two western sites, E.C. Manning park and Smithers B.C., Canada, and our two eastern sites, Harvard Forest, MA, USA and St. Hippolyte, QC, Canada. Weather data for the fall of 2019 and winter 2020 was obtained from the Hope Slide weather station for our E.C. Manning park sites and the Smithers airport weather station for our Smithers site (Environment and Climate Change Canada, 2026). For our eastern sites, weather data for the fall of 2014 and winter 2015 was obtained from weather stations at Harvard Forest (Boose, E and Gould, E and VanScoy, M, 2026) and the St. Hippolyte weather station (Environment and Climate Change Canada, 2026).

Site	Chilling treatment	Chill hours	Utah model	Chill portions
Harvard forest	Field chilling	892	814.50	56.62
Harvard forest	Field chilling + 30 d at 4°C	2140	2062.50	94.06
St. Hippoltye	Field chilling	682	599.50	44.63
St. Hippoltye	Field chilling + 30 d at 4°C	1930	1847.50	82.06
Smithers	Field chilling + 21 d at 4°C	1317	1368.00	54.95
Smithers	Field chilling + 56 d at 4°C	1965	2016.00	74.67
Manning park	Field chilling + 21 d at 4°C	1213	1377.00	55.09
Manning park	Field chilling + 56 d at 4°C	1861	2025.00	75.33

Table S3: Summary output from a phylogenetic Bayesian model in which species are partially pooled and phylogeny is included on the intercept. See the statistical analysis section of the methods for more detail.

	mean	5%	25%	75%	95%
α_{sp}	30.41	25.96	28.55	32.15	35.23
λ	0.43	0.08	0.27	0.59	0.75
$\beta_{forcing}$	-10.62	-11.90	-11.17	-10.09	-9.34
$\beta_{photoperiod}$	-3.36	-3.97	-3.61	-3.10	-2.73
$\beta_{chilling}$	-14.81	-16.69	-15.60	-14.03	-12.91
$\beta_{Manning\ park}$	2.09	1.45	1.82	2.35	2.74
$\beta_{Harvard\ forest}$	-6.62	-8.38	-7.30	-5.91	-4.93
$\beta_{St.\ Hippolyte}$	-8.76	-10.41	-9.43	-8.10	-7.19
$\beta_{forcing \times photoperiod}$	0.83	-0.46	0.31	1.35	2.13
$\beta_{forcing \times chilling}$	8.37	6.98	7.79	8.93	9.79
$\beta_{photoperiod \times chilling}$	-0.63	-2.00	-1.18	-0.07	0.75
$\beta_{photoperiod \times Manning\ park}$	0.55	-0.77	0.01	1.07	1.84
$\beta_{chilling \times Manning\ park}$	-0.35	-3.05	-1.37	0.74	2.13
$\beta_{forcing \times Harvard\ forest}$	3.79	1.73	2.94	4.65	5.87
$\beta_{photoperiod \times Harvard\ forest}$	-1.56	-2.98	-2.14	-0.97	-0.11
$\beta_{chilling \times Harvard\ forest}$	8.59	5.61	7.43	9.77	11.55
$\beta_{forcing \times St.\ Hippolyte}$	4.86	2.83	4.07	5.67	6.85
$\beta_{photoperiod \times St.\ Hippolyte}$	-2.15	-3.46	-2.70	-1.60	-0.82
$\beta_{chilling \times St.\ Hippolyte}$	7.18	4.65	6.18	8.22	9.68

Table S4: Summary output from a phylogenetic Bayesian model in which the phylogeny is included on the residuals. See the statistical analysis section of the methods for more detail.

	mean	5%	25%	75%	95%
α_{sp}	28.90	26.20	27.80	29.98	31.62
λ	0.49	0.05	0.25	0.75	0.96
$\beta_{forcing}$	-10.60	-11.90	-11.14	-10.12	-9.35
$\beta_{photoperiod}$	-3.30	-3.99	-3.60	-3.08	-2.72
$\beta_{chilling}$	-14.80	-16.72	-15.58	-14.04	-12.96
$\beta_{Manning\ park}$	2.10	1.45	1.84	2.34	2.73
$\beta_{Harvard\ forest}$	-5.90	-7.57	-6.58	-5.26	-4.30
$\beta_{St.\ Hippolyte}$	-8.10	-9.68	-8.75	-7.50	-6.59
$\beta_{forcing \times photoperiod}$	0.90	-0.42	0.32	1.39	2.12
$\beta_{forcing \times chilling}$	8.40	6.96	7.80	8.95	9.82
$\beta_{photoperiod \times chilling}$	-0.70	-2.15	-1.26	-0.09	0.78
$\beta_{forcing \times Manning\ park}$	-1.70	-3.11	-2.29	-1.20	-0.35
$\beta_{photoperiod \times Manning\ park}$	0.60	-0.75	0.05	1.11	1.84
$\beta_{chilling \times Manning\ park}$	-0.30	-2.94	-1.32	0.68	2.22
$\beta_{forcing \times Harvard\ forest}$	3.80	1.76	2.95	4.66	5.92
$\beta_{photoperiod \times Harvard\ forest}$	-1.60	-3.05	-2.15	-0.98	-0.14
$\beta_{chilling \times Harvard\ forest}$	8.80	5.89	7.62	10.01	11.79
$\beta_{forcing \times St.\ Hippolyte}$	4.90	2.86	4.03	5.67	6.88
$\beta_{photoperiod \times St.\ Hippolyte}$	-2.20	-3.53	-2.71	-1.60	-0.79
$\beta_{chilling \times St.\ Hippolyte}$	7.40	4.92	6.40	8.35	9.78

Table S5: Proportion of surviving samples per species for which budburst was observed in our controlled environment study of western sites. See (Flynn and Wolkovich, 2018) for survival in our study of eastern sites.

Species name	Proportion budburst	Plant type
Acer glabrum	0.83	tree
Alnus incana	1.00	shrub
Alnus viridis	0.92	shrub
Amelanchier alnifolia	0.99	shrub
Betula papyrifera	1.00	tree
Cornus stolonifera	0.99	shrub
Lonicera involucrata	0.87	shrub
Menziesia ferruginea	0.80	shrub
Populus balsamifera	0.98	tree
Populus tremuloides	0.90	tree
Rhododendron albiflorum	1.00	shrub
Ribes lacustre	0.82	shrub
Rubus parviflorus	0.94	shrub
Sambucus racemosa	0.95	shrub
Shepherdia canadensis	1.00	shrub
Sorbus scopulina	0.99	shrub
Spiraea betulifolia	0.94	shrub
Spiraea pyramidata	0.92	shrub
Symphoricarpos albus	0.84	shrub
Vaccinium membranaceum	0.90	shrub
Viburnum edule	1.00	shrub

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